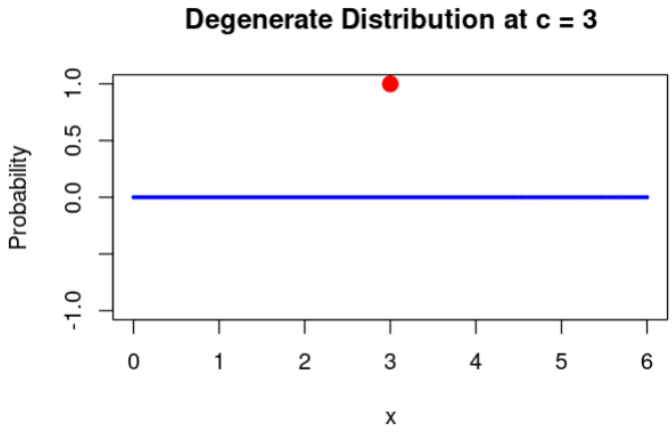
Degenerate Distribution

1 Definition

A random variable X is said to have Degenerate Distribution if $P(X = c) = 1$ for some c.

PMF of X is $f_X(x) = \begin{cases} 1 & x = c \\ 0 & \text{otherwise} \end{cases}$



CDF of X is $F_X(x) = P(X \leqslant x) = \begin{cases} 0 & x < c \\ 1 & x \geqslant c \end{cases}$

2 Moments

$$\begin{aligned} \mu'_r = E(X^r) &= c^r P(X > c) \\ &= c^r \end{aligned}$$

$$\mu'_1 = E(X) = c$$

$$\begin{aligned} V(X) &= \mu'_2 - \mu'^2_1 \\ &= c^2 - c^2 \\ &= 0 \quad \text{(It has no variability)} \end{aligned}$$

Note that $V(X) = 0$ iff $E(X-\mu)^2 = 0, \mu = E(X)$
iff $P(X - \mu = 0) = 1$
iff X is a degenerate random variable.